Topic: Hierarchical Retail Sales Forecasting and Error Analysis: A Baseline Comparison Approach

1. The Practical or research question
How can store managers predict daily sales for the next 28 days to avoid stockouts and overstocking?

2. The type of statistical task
This is a multivariate time series forecasting task, which also involves hierarchical forecasting. The core task is to use historical sales data and relevant covariates (e.g., prices and calendar events) to perform point forecasting for future continuous numerical values (sales volume). Model performance will be evaluated using a strict temporal train–validation–test split, comparing a simple historical moving average baseline with a multi-feature candidate model using RMSE and MAE.

3. The proposed dataset
We will use the official dataset from the Kaggle M5 Forecasting - Accuracy competition. (Link: https://www.kaggle.com/competitions/m5-forecasting-accuracy/data)

The dataset combines 4 primary relational files:

calendar.csv (contains information about the dates on which the products are sold);
sales_train_validation.csv (contains the historical daily unit sales data per product and store [d_1 - d_1913]);
sell_prices.csv (contains information about the price of the products sold per store and date);
sales_train_evaluation.csv (includes sales [d_1 - d_1941] (labels used for the Public leaderboard)).

To ensure computational feasibility and reproducibility within project constraints, we will model a representative subset of product categories or stores.

4. The relevant population and unit of analysis
- Relevant Population: Daily unit sales demand for consumer goods across Walmart stores in California, Texas, and Wisconsin (specifically focusing on the selected representative store or product category sub-population for computational feasibility).
- Unit of Analysis: Each row in the dataset represents the daily unit sales volume (Y) of a specific item in a specific store on a specific date.

5. The intended outcome or target
- Primary Outcome: Minimize daily sales forecasting errors—specifically by minimizing Root Mean Squared Error (RMSE) and Mean Absolute Error (MAE).
- Secondary Outcome: Identify specific scenarios where forecasting errors spike (such as items with frequent zero-sales days) to establish clear rules for when human intervention is required.

6. Foreseeable sampling, measurement, missing-data, privacy, licensing, or access limitations
- Zero-inflation and intermittent demand: Many items have zero sales on many days, creating a "zero-inflated" distribution that is difficult for standard regression models to fit.
- Censored data (Measurement bias): Zero sales may indicate either "no demand" or "out of stock." Without inventory level data, it is difficult to distinguish between these two, leading to measurement bias.
- External factors (Unobserved covariates): Unforeseen events (e.g., extreme weather, local pandemics) not included in the dataset may act as confounding variables, affecting the accuracy of the forecast.
- Computational complexity: The high dimensionality of the data (millions of observations) requires significant computational resources for model training and hyperparameter tuning, which we mitigate by modeling a representative subset of product categories or stores.
- Privacy and Licensing: The dataset is publicly accessible on Kaggle under standard competition terms. It contains aggregated store-level sales and no personally identifiable information (PII), posing zero privacy or licensing restrictions.

7. The Evidence Needed to Support the Intended Claim or Decision
To support the claim that the proposed model improves inventory efficiency, we need:

- Accuracy metrics: Comparison of RMSE and MAE against a simple Historical Moving Average baseline model to prove statistically significant error reduction.
- Subgroup & category error analysis: Evidence of prediction performance across different product categories and store locations to identify where the model excels and where it struggles.
- Feature contribution (Ablation): Evidence showing the relative contribution of key covariates (e.g., whether adding promotional events or price lag features noticeably reduces forecasting errors for specific product categories).
- Residual & uncertainty analysis: Plots showing that residuals are randomly distributed and diagnostics identifying specific edge cases (e.g., intermittent zero-sales days) where model forecasts should not be trusted without human intervention.